



RenderRank

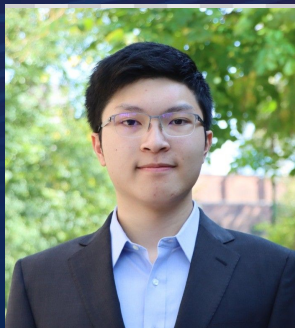
Cross-platform 3D rendering GPU benchmarks in the browser



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Team Profile



Charles Jin

Senior, Class of 2027
Major: CIS



Eric Sun

Senior, Class of 2027
Major: Physics, CIS



Michael Ren

Senior, Class of 2027
Major: AI, EE



Stan Chen

Senior, Class of 2027
Major: CIS, ROBO



Yiding Tian

Senior, Class of 2027
Major: CMPE, CGGT

GPU owners need an easy way to compare their devices on the same configurable rendering workloads.

Problems to be solved:



Access friction

Native benchmark apps add an installation step on each device.



Comparison friction

Different settings or environments can make scores hard to interpret.

Current approaches and limitations

Existing tools cover parts of the need. Our focus is configurable WebGPU rendering with a consistent test protocol.

Approach	What it offers	Gap relative to RenderRank
3DMark	Rich native graphics tests, including cross-platform tests	Requires a native app. Platform support varies by test.
Basemark Web 3.0	Browser benchmark with WebGL graphics and broader web tests	Doesn't actually do any visual benchmarks, only simulates GPU load

Proposed distinction: adjustable scene features, repeatable presets, and personal result history in one browser workflow.

Proposed approach

Choose a scene

Adjust crowds, lighting, post-processing, and screen space effects.

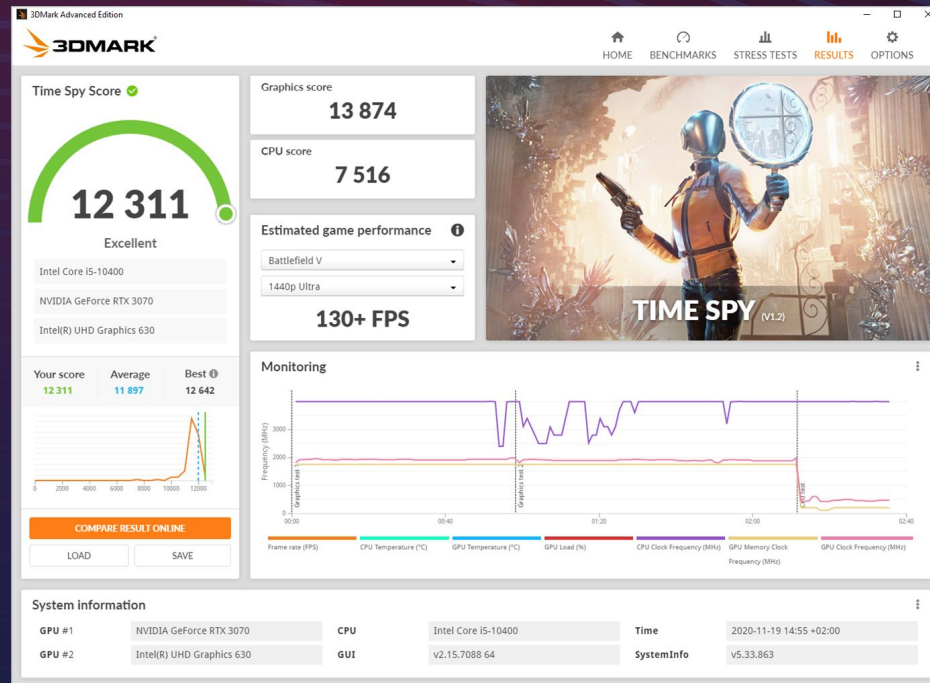
Run a controlled test

Track FPS, frame times, and rendering throughput.

Save and compare

Review your devices. Submit to the public leaderboard only with consent.

Ranked runs use locked presets. Custom runs support exploration.



Requires a WebGPU-compatible browser and GPU.

Stretch goal: diffusion image-generation benchmark

Technical merits

01

Clustered lighting

Use GPU compute to assign lights to 3D view clusters, then shade with each cluster's light list.

02

Browser rooted

No need to download software or setup environment for benchmarking. All done within one click

03

Repeatable measurement

Fix the scene seed, camera path, resolution, and version. Warm up, repeat runs, and report variation.

04

Comparable results

Separate display FPS from completed-work throughput. Treat optional GPU timestamps as diagnostics.

Broader impacts



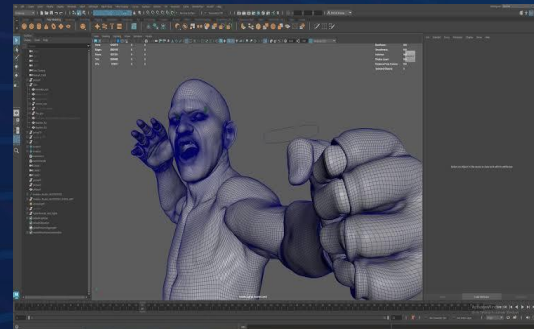
Device owners

Compare hardware you already own and see which rendering features affect performance.



Web developers

Explore rendering bottlenecks on varied devices and choose suitable quality settings.



3D Artists

Use interactive scenes to check GPU performance of 3D scenes without coding knowledge.

Responsible design: opt-in sharing, minimal device metadata, and results grouped by test version and environment.

Milestones and proposed timeline

Proposed Fall 2026 checkpoints

Late Sep

Next milestone

One scene with repeatable runs and CSV export.

Demo on two devices.

Late Oct

Integrated alpha

Three scene presets, saved history, and a leaderboard prototype.

Test supported OS/browser pairs.

Early Dec

Semester deliverable

Deploy the rendering suite. Validate repeatability and usability.

Publish methodology and demo.

Proposed division of work

Each teammate owns code and a reviewable deliverable for early Dec.

Owner	Implementation responsibility	Deliverable by December
Charles Jin	Benchmark runner and metrics	Fixed-run protocol and CSV export
Eric Sun	Frontend and user experience	Preset controls and live metrics UI
Michael Ren	API, storage, and leaderboard	Result schema and save/load endpoint
Stan Chen	Scenes, effects, and user research	One configurable scene and 3 user interviews
Yiding Tian	WebGPU renderer and GPU algorithms	Standardized test procedure and scene

Shared responsibility: code reviews, device testing, integration, and presentation preparation

The background is a dark blue gradient with subtle, wavy lines. Scattered around the central text are several 3D geometric objects: a yellow torus at the top left, a blue curved tube on the left, a purple triangular prism at the top right, an orange sphere at the bottom right, an orange wavy line with an arrowhead at the bottom left, and a pink wavy line at the bottom center. All these objects have soft shadows cast beneath them.

Thank you

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01 About us

You can enter a subtitle here if you need it

The slide title goes here!

Do you know what helps you make your point clear? Lists like this one:

- They're simple
- You can organize your ideas clearly
- You'll never forget to buy milk!

And the most important thing: the audience won't miss the point of your presentation



What sets us apart?



Mercury

Mercury is the closest planet to the Sun and the smallest one in the Solar System—it's only a bit larger than the Moon



Venus

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot, even hotter than planet Mercury

What do we offer?



Mercury

It's the closest planet to the Sun and the smallest in the Solar System



Venus

Venus has a beautiful name and is the second planet from the Sun



Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

A picture always the reinforces the concept

Images reveal large amounts of data, so remember: use an image instead of a long text. Your audience will appreciate it



Target audience worldwide



Mars



Despite being red, Mars is actually a cold place

Mercury



Mercury is the smallest planet in the Solar System

Neptune



Neptune is the farthest planet from the Sun

You can use this graph



Venus

Venus is the second planet from the Sun



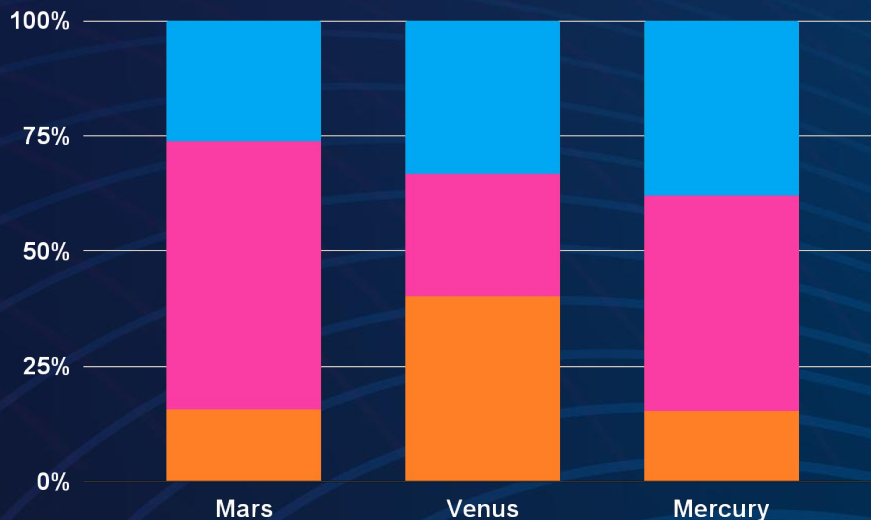
Mercury

Mercury is the closest planet to the Sun



Mars

Despite being red, Mars is a very cold place



Follow the link in the graph to modify its data and then paste the new one here. **For more info, click here**

Know the process

Step 1
Mars is actually a
very cold place

Step 4
Mercury is the
smallest planet



Step 2
Neptune is very far
away from us

Step 3
Saturn is a gas giant
and has rings

This is a table



	Mass	Diameter	Surface gravity
Mercury	0.06	0.36	0.36
Mars	0.11	0.53	0.38
Saturn	95.2	9.4	1.16



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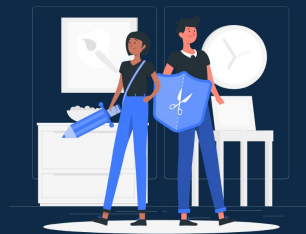
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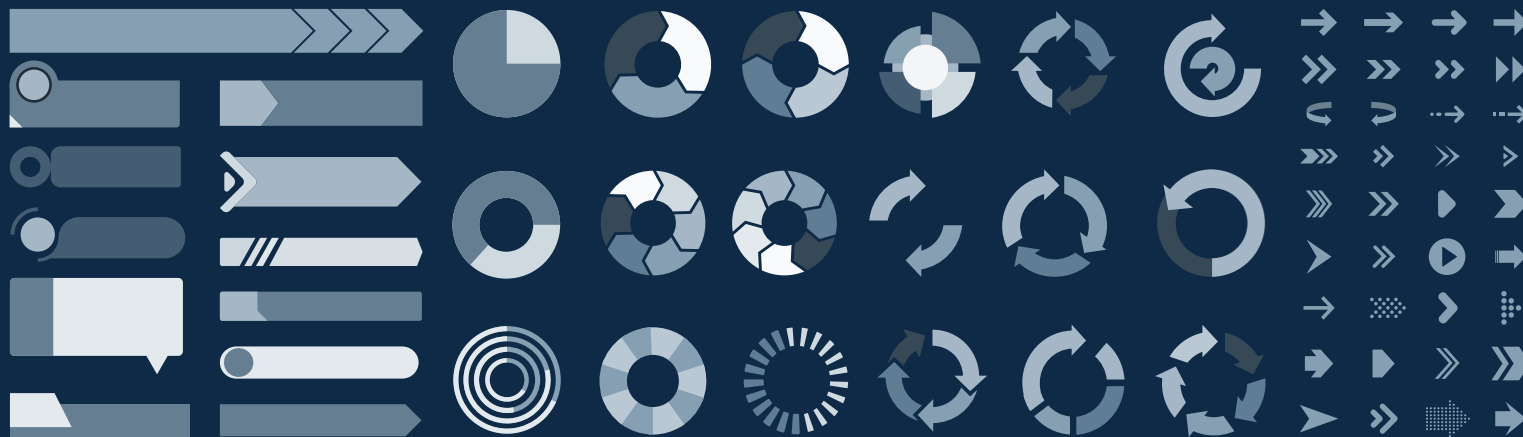


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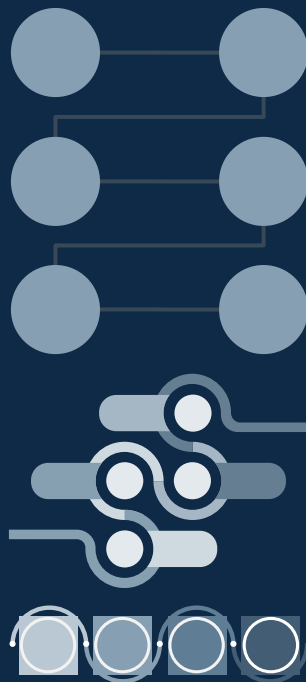
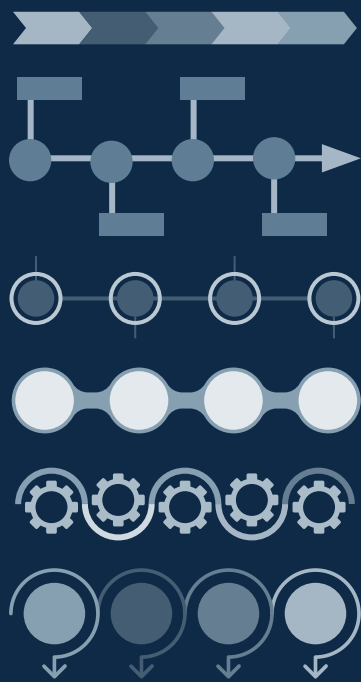
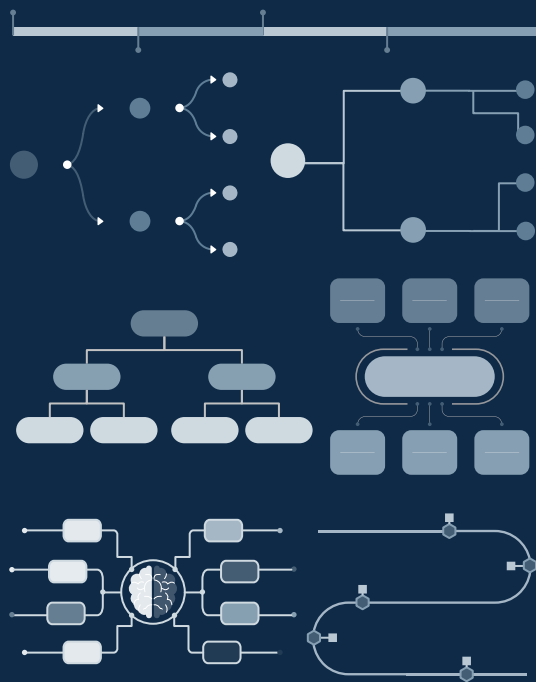
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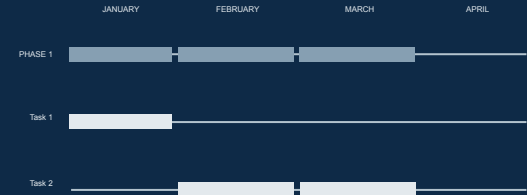
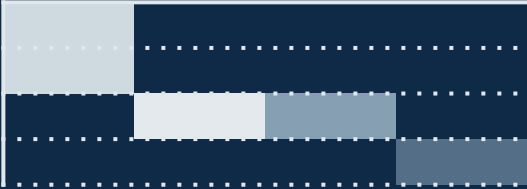
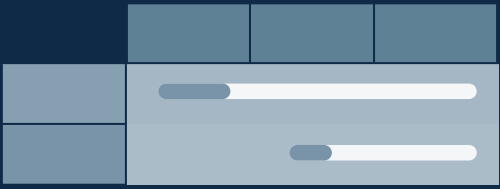
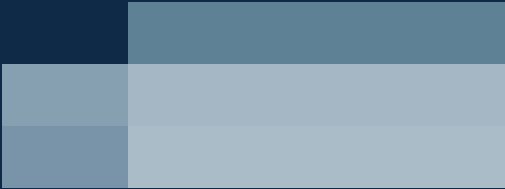
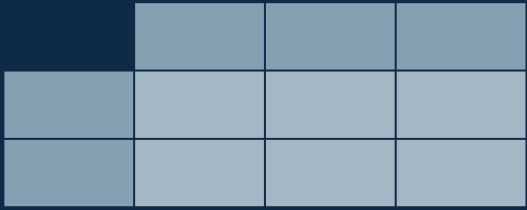
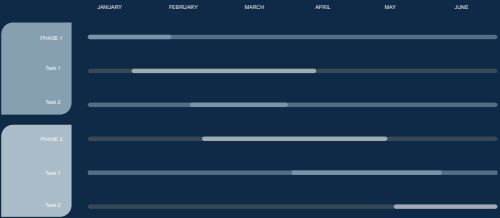
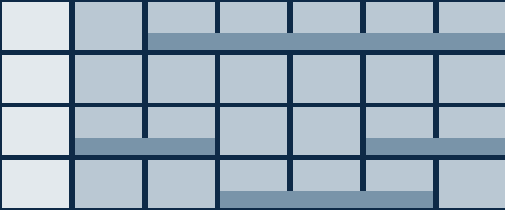
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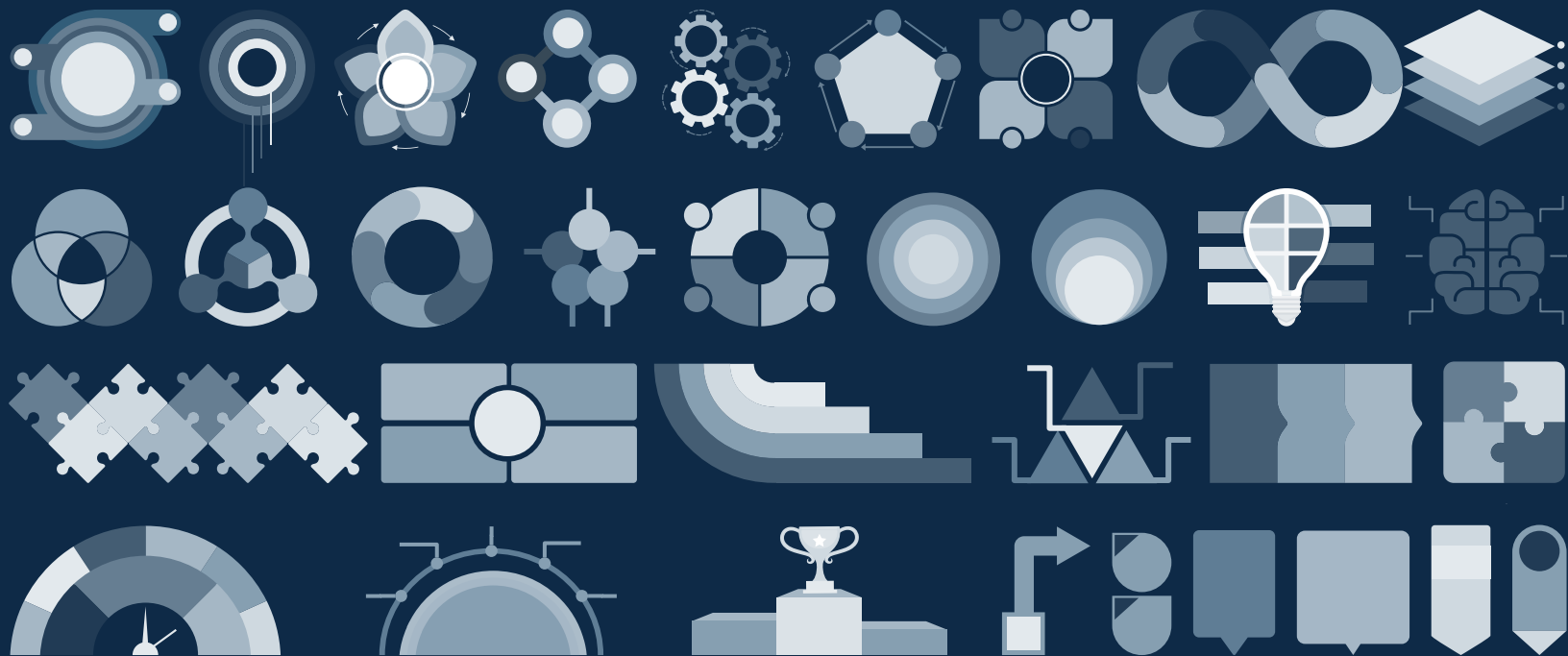
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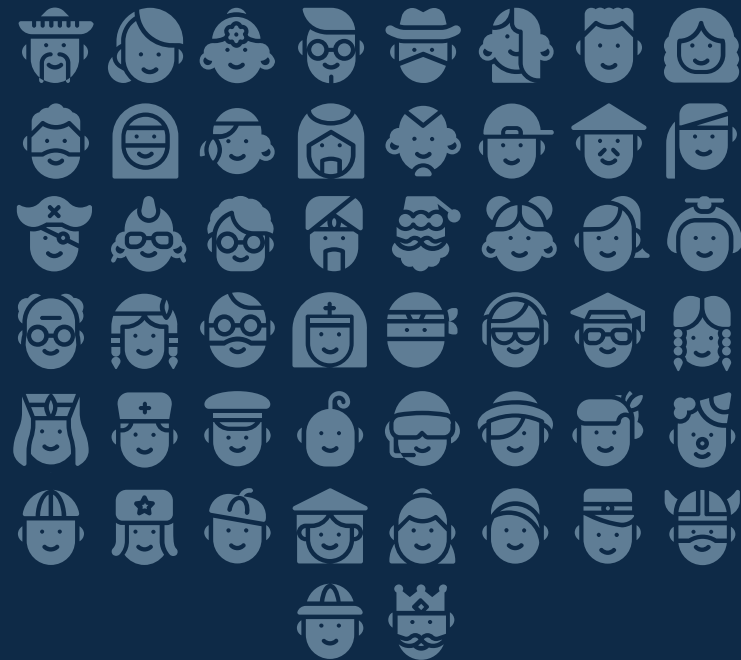
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